



SCIENCE OLYMPIAD
— AT THE —
UNIVERSITY OF FLORIDA

Northern Regional: January 19th, 2019

Water Quality C Test

DO NOT WRITE ON THIS TEST

Water Quality
Division C

This test will consist of 3 sections and 9 total stations. Stations 1, 2, 3, and 4 cover information about section 1, freshwater ecology (80 points). Stations 5, 6, and 7 will cover section 2, aquatic flora and fauna (74 points). Stations 8 and 9 will cover the chemistry and analysis aspects of water quality, section 3 (31 points).

Name(s): _____

Points: ____ / 185

Section 1 Freshwater Ecology

Station 1 (20 Points)

1. Which of these freshwater nutrients does not commonly have a gaseous phase of its nutrient cycle? (1)

- A) Nitrogen
- B) Phosphorous
- C) Oxygen
- D) Carbon

2. What is the term for smaller particles that bind, mostly due to chemicals added to a mixture, to form larger particles? (2)

3. Match each of the following water treatment steps to its description. Letters may be used only once, match only one letter to each description. (5)

- A) Filtration
- B) Disinfection
- C) Coagulation
- D) Sedimentation
- E) Reverse Osmosis

- 1. Water passes through various porous layers such as sand to remove dissolved particles.
- 2. An additional filter that uses a semi-permeable membrane to remove any remaining impurities.
- 3. An antiseptic is added to protect the water from germs.
- 4. Heavy particles settle to the bottom of the water supply and are filtered out.
- 5. Ions are added to the water, which bond to oppositely charged ions, forming particles that can be filtered out more easily.

4. List four things that stormwater runoff transports that can be harmful in freshwater ecosystems. (4)

5. Which of the following is an autotroph in freshwater ecosystems? (1)

- A) Zooplankton
- B) Salmon
- C) Cattails

D) Tadpoles

6. What is the term for an excess of nutrients in a freshwater ecosystem? (2)
7. List four freshwater ecosystem autotrophs. (4)
8. Which of the following is **false** about freshwater ecosystems? (1)
 - A) Freshwater ecosystems have low salt concentrations, with less than one gram per liter of dissolved solids.
 - B) Freshwater ecosystems have higher oxygen concentrations
 - C) Sulfur and Nitrogen Oxide abundance in freshwater can cause acid rain, causing bodies of freshwater to become acidic.
 - D) The different types of freshwater biomes include ponds and lakes, wetlands, streams and rivers, and estuaries.

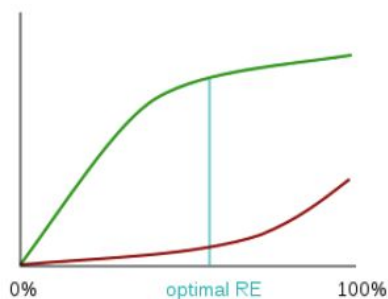
Station 2 (25 Points)

1. What are the four main features that define the ecology of streams and rivers? (4)
2. Of the 20,000 species of fish, Biologists group them into three main groups: Agnatha, Chondrichthyes, and the Osteichthyes. Match each species with its main group. (5)
 - A: Shark
 - B: Ray
 - C: Salmon
 - D: Trout
 - E: Lamprey
3. True/False: Small bodies of water like puddles and ditches are considered freshwater biomes. (1)
4. Freshwater biomes supply how much of the world's freshwater? (1)
 - A. 10%
 - B. 33%
 - C. 50%
 - D. 25%
 - E. 90%
5. What are four characteristics that describe invasive plants? (4)
6. Of the following, which characteristics are found in the phylum annelida? (3)
 - A: segmented
 - B: stiff jointed appendages
 - C: sexual and asexual reproduction
 - D: sponges
 - E: leeches
7. What is the order of each of the following species? (4)
 - A. Mayfly
 - B. Stonefly
 - C. Dragonfly
 - D. Caddisfly
8. Freshwater ecosystems are divided into ____ and ____ systems. (2)

9. Increased levels of algae blooms result in which of the following? (1)

- A. acid rain
- B. lower levels of nutrients in the water
- C. thermal pollution
- D. increased levels of eutrophication
- E. seasonal turnover

Station 3 (20 Points)



1. Refer to the above graph.
 - a. What type of reproduction is modeled by the graph? (1)
 - b. Label the x-axes and each of the lines. What determines the “optimal RE”? (3)
 - c. (True/False) This type of reproduction is consistent with that of a salmon. (1)
 - d. Is this type of reproduction consistent with r or K selection? (1)

Community	Species 1	Species 2	Species 3	Species 4	Species 5
<i>x</i>	3	4	2	0	0
<i>y</i>	3	3	0	4	1

2. Refer to the above chart. A researcher visits two communities and collects data regarding the populations of five different species in the two communities.
 - a. What is Sorensen’s coefficient of similarity for these two communities? (2)
 - b. What are the maximum and minimum values of Sorensen’s coefficient? What do they signify? (2)
 - c. Calculate Simpson’s index of diversity for both communities. Is it appropriate to compare the two values, and why? (3)



3. Refer to the river in the above image.
 - a. Identify the type of this river. (1)
 - b. Explain the cause of the color of the river. (3)
 - c. Compared to “regular” rivers, the water in this river has a (higher / lower) pH, (more / less) phosphorus content, and (higher / lower) levels of dissolved solids. (3)

Station 4 (15 Points)

1. The following questions relate to acid rain
 - a. Why do lakes have a natural buffering capacity? Write the relevant chemical equation. (2)
 - b. Write the two relevant chemical reactions when a strong acid reacts with the buffer. (1)
 - c. Write the chemical reaction for the production of carbonic acid in water. (1)
 - d. If K_{a1} of H_2CO_3 is 4.3×10^{-7} and K_{a2} of H_2CO_3 is 4.8×10^{-11} , calculate the pH of a solution of 1M H_2CO_3 . Assume nothing else is present in the solution. Hint: to solve this problem quickly, it is necessary to make assumptions. (5)
2. Define the following processes of the water cycle. (3)
 - a. Precipitation intercepted by foliage that is evaporated
 - b. Movement of water through the atmosphere
 - c. Flow of water in the vadose zone and aquifers
3. Which of the following is/are true of lake stratification? (3)
 - a. Aeration can be used to lessen thermal stratification
 - b. The thermocline lies between the metalimnion and the hypolimnion
 - c. Only dimictic and polymictic lakes exhibit mixing (turning over) of water
 - d. Overturn in dimictic lakes occurs twice a year in the winter and summer
 - e. Lake stratification can affect the quality of drinking water
 - f. Carbon dioxide accumulating in meromictic lakes can result in a limnic eruption

Section 2 Aquatic Flora and Fauna

Station 5 (27 Points)

Match the freshwater habitat to its description. Letters may be used once, more than once, or not at all, but please only pick one letter per description. (6)

- A) Edgewater
- B) Riffles
- C) Runs
- D) Pools

1. Includes overhanging vegetation from banks
2. High in oxygen and especially rich in insect diversity
3. Slow moving water at edges or behind rocks and logs
4. Holding spot for amphipods and isopods
5. Home of fast moving bugs and beetles
6. Water tumbling over rocks

Questions 7-10. Of the given aquatic invertebrates, which is least sensitive to pollution? (4)

7.

- A) water mite
- B) leech
- C) scud
- D) air breathing snail

8.

- A) tubifex
- B) water scorpion
- C) caddisfly
- D) midge

9.

- A) blood midge
- B) flatworm
- C) dobsonfly
- D) riffle beetle

10.

- A) aquatic sowbug
- B) gilled snails
- C) water penny
- D) deer/horse fly

Questions 11-23. The following is a list of aquatic insects. Mark A if they undergo complete metamorphosis and mark B if they undergo incomplete metamorphosis. (13)

- 11. Damselflies
- 12. Dragonflies
- 13. Alderflies
- 14. Dobsonflies
- 15. Stoneflies
- 16. Mayflies
- 17. Beetles
- 18. Moths
- 19. Waterbugs
- 20. Caddisflies
- 21. Backswimmer
- 22. True flies
- 23. Waterboatman

Match the habitat to the feeding type. (4)

- A) Collectors
- B) Shredders
- C) Scrapers
- D) Predators

- 24. All types
- 25. Areas with enough light for algae growth
- 26. Stream bottom
- 27. Areas with tree cover

Station 6 (16)

1. List the environmental effects caused by aquatic nuisance species. (6)
2. List the economic impacts caused by aquatic nuisance species. (6)
3. List the public health impacts caused by aquatic nuisance species. (4)

Station 7 (31 Points)

A



B



C



D



E



F



1. Refer to organism A.
 - a. What is the common name of this organism? (1)
 - b. What is the lifespan of these adult organisms? (3)
 - c. The larvae of this organism are some of the (largest/smallest). (1)
2. Refer to organism B.
 - a. Identify the phylum this organism belongs to. (1)
 - b. What is the feeding strategy of this organism? (2)
 - c. What is the reproductive strategy of this organism? (2)
3. Refer to organism C.
 - a. What organism is this? (1)
 - b. What are the nicknames of the larvae of this organism? (2)
 - c. What are differences between the males and females in this species? (2)

4. Refer to organism D.
 - a. What is the scientific name of this organism? (1)
 - b. What is the trophic level of this organism? (1)
 - c. What is the primary prey of this organism? (1)
 - d. What are the negative impacts of this organism? (2)

5. Refer to organism E.
 - a. What is the scientific name of this organism? (1)
 - b. How do they attach to various surfaces? (1)
 - c. Why is their colonization harmful? (2)
 - d. What is the feeding mechanism of this organism? How is this both beneficial and harmful? (3)

6. Refer to organism F
 - a. This organism was found in the Northeast. What is most likely the genera of this organism? (1)
 - b. Why is this organism an indicator of good water quality? (2)
 - c. What is the approximate size of this organism? (1).

Section 3 Water Quality Chemistry and Analysis

Station 8 (11 Points)

Questions 1-4 test your knowledge on different chemical analysis tests for water quality.

1. Temperature affects the amount of gases such as oxygen that can be dissolved in the water. Cold water holds (more/less) oxygen than warm water. Temperature (increases/decreases) metabolic rate of aquatic organisms. Raising water temperature (increases/decreases) decomposition rate of organic matter. (3)
2. pH is measured with a pH _____. Normal pH of rivers in the US is between 6.5 and _____. Acid rain has a pH of less than _____. (3)
3. _____ measures the clarity or haziness of water in a given body. (1)
4. Identify the most appropriate tool to measure and analyze each of the following indicators.
 - a. Pesticides and organic pollutants (1)
 - b. Heavy metals (1)
 - c. Total dissolved solids (1)
5. Which of these common dissolved solids are not essential nutrients for life. (1)
 - A) magnesium
 - B) nitrogen
 - C) phosphorus
 - D) sulfur

Station 9 (20 Points)

1. Identify the aquatic life that can survive at the lowest pH levels. (1)
A) snails, clams
B) carp, catfish
C) algal or rooted plants
D) crustaceans
E) mayfly, dragonfly nymph
2. Increased bacteria, decreased oxygen levels, and death of aquatic life are effects of pollution most likely caused by (be specific!): (3)
3. Identify natural factors that may affect dissolved oxygen levels. (8)
4. Identify human factors that may affect dissolved oxygen levels. (4)
5. Identify appropriate methods or remedies to remove radioactive, chemical, and/or biological pollutants from water. (4)